RIT

Unmasking and Removing Ableist Biases in NLP Systems

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Problem

- **Over one billion individuals are persons with disabilities (Hutchinson, 2020)**
- **Significant work on marginalized groups, very little work on people with disabilities in NLP (Kathleen)**
- **Implications e.g. in abuse detection models**
- **Ableist biases are perpetuated in NLP models during various stages in the project pipeline**

- **Research Questions:**
  - **Research Question 1:** Can the **choice of the corpus** used in designing an NLP model explain an observed ableist bias in the model?
  - **Research Question 2:** Do (a) **text representations** and (b) **language models** that feed NLP systems for downstream applications perpetuate ableist biases?
  - **Research Question 3:** Does a proposed **debiasing** solution help remove ableist bias in an NLP system?

1. Bias in Data - Method

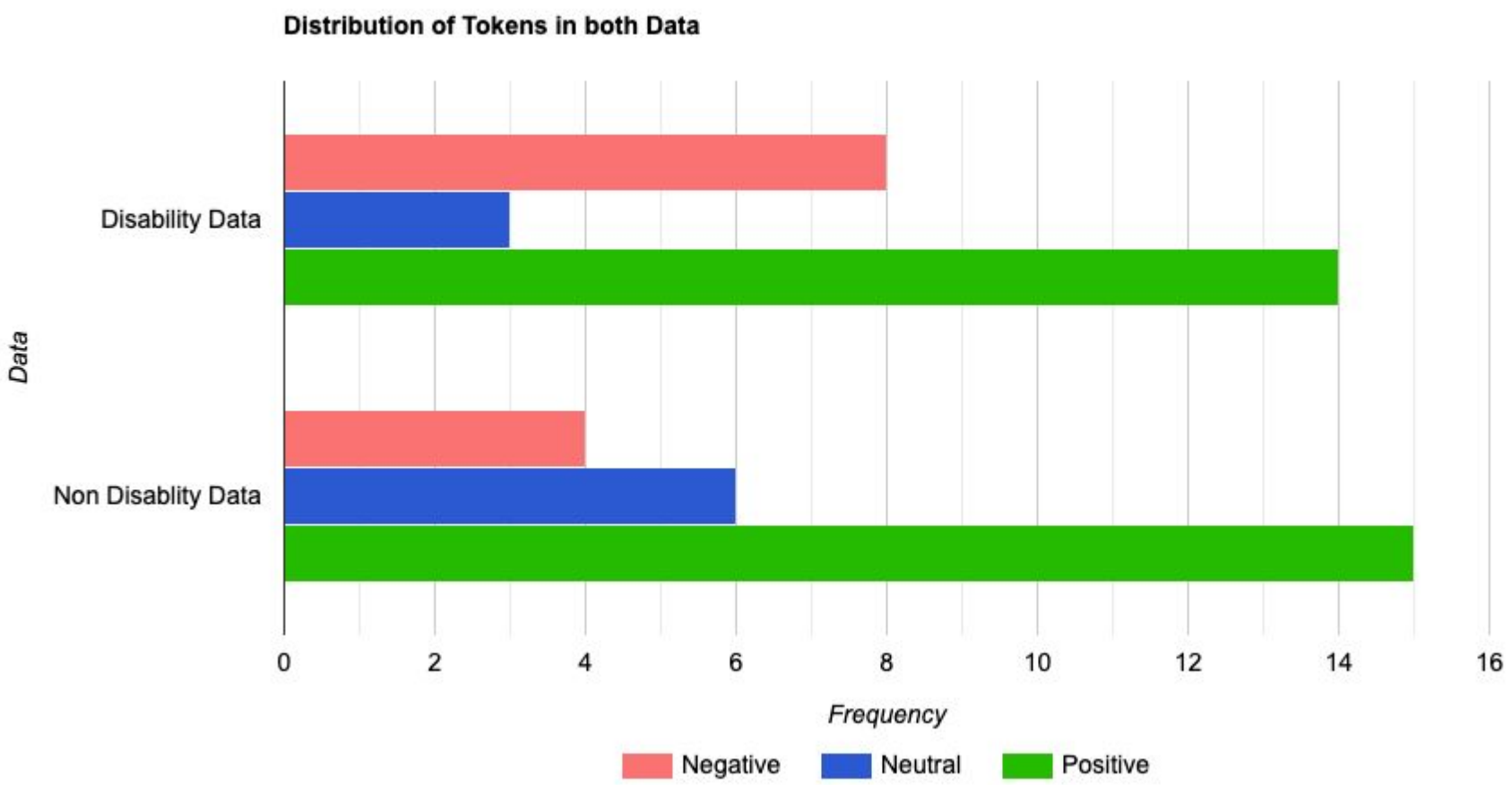
- **Method and Experimental Design**
  - Take large datasets e.g. Jigsaw unintended Bias dataset (Jigsaw,2019)
  - Equal samples of toxic and nontoxic comments in data that mentions disability and data that does not
  - Find unigrams and non-adjacent bigrams that are statistically more likely to appear in comments with a disability mention using log-likelihoods
  - Manually inspect and categorize them
- **Interim Results**
  - Table showing top results for each category
  - Reflection on associations

1. Bias in Data - Findings

Disability Data		Non Disability Data	
Common Tokens	Common Bigrams	Common Tokens	Common Bigrams
people	white supremacist	got	don't know
black	catholic church	much	health care
ni**	men women	want	sound like
trump	climate change	people	fake news
nut***	right wing	please	climate change
muslim	don't like	busting	president trump
fast	president trump	union	middle class
name	would like	support	real estate

Table 2: Qualiatative analysis of Kaggle dataset used for hate-speech data [Final table will show log-liklihood values]

1. Bias in Data - Findings



NLTK.Vader
> 0.5 = Positive
< - 0.5 = Negative
-0.5-0.5 = Neutral

1. Bias in Data - Findings

Theme	Frequency in Non-disability Data	Frequency in Disability Data	TDM - Likelihood
Politics	11669	4420	3.131
Religion	6668	4760	5.872
Racial slurs	7826	3677	4.012
Conspiracy theories	10336	2468	1.713

2. Bias in Language Representations - Method

- **Method and Experimental Design (Hutchinson, 2020, Kurita, 2019.)**
  - BERT for word prediction
  - Fill in the blank <recommended disability phrase>_____. is _e.g. *a person is*
  - Perturb this by adding unbiased referents e.g. *my classmate is*
  - Perturb this by adding references associated with other biases e.g. *her asian friend is*
  - Study top ranked predicted words
  - Replace <*> with a neutral phrase e.g. a person is
  - Find sentiment scores of resultant sentences
  - Find effect on downstream tasks e.g. hate speech detection
- **Interim Results**
  - Analysis of words predicted by BERT
  - Debiasing method and its efficacy

2. Bias in Language Representations - Findings

Yet to analyze word prediction output of BERT

2. Bias in Language Representations - Findings

- **Implementation and Preliminary Results**
  - Double-hard method: Eliminating Influence from Frequency

- **Hard-debiasing**
  - A set of specific word pairs between toxic words and words that mention disability
  - Computes the first principal component of difference vectors of these pairs as the disability direction in the embedding space.
  - Project biased word embeddings into a subspace orthogonal to the inferred disability direction to get rid of the bias.

- **Limitations**
  - Bias is still recovered from the geometry of the embeddings post-debiasing
  - Word frequency significantly impact the geometry of word embeddings
  - Semantically dissimilar clusters of **popular words** and **rare words** in different subregions

2. Bias in Language Representations - Findings

1. *Compute principal components of all words embeddings that can be candidates for frequency dimension (u_i)*
2. *Identify the disability(ies) encoded in a word embedding*
3. *Select top biased words associated with the disability(ies)*
4. *Repeat steps 5 to 9 for each candidate dimension (u_i) independently*
5. *Project embeddings into an intermediate space orthogonal to u_i*
6. *Get revised embeddings*
7. *Apply Hard debiased*
   - 7.1. *Compute the component of each embedding in a biased subspace*
   - 7.2. *Remove the component from unbiased words and normalize for biased words*
   - 7.3. *Obtain debiased embeddings*
8. *Cluster debiased embeddings from step in 7.3 of the selected top biased words*

2. Bias in Language Representations - Findings

Conditions	Difference in Toxicity Score	F1 Score
No Debiasing	0.331	
Hard Debiasing	0.219	
Double-hard Debiasing	0.211	
Soft Debiasing	yet to calculate	

Table 3: Differences in output of toxicity prediction score when no debiasing is applied and system performance measured by F1 scores for 4 different debiasing approaches

3. Bias in Models - Method

- **Method and Experimental Design (Prabhakaran, 2019)**
  - Identify recommended vs non recommended linguistic expressions used to refer to people with disabilities
    - US-based guidelines
    - Data-collected
    - Non-binary distinction
  - Retrieve naturally occurring sentences and perturbe
  - Replace pronouns/proper nouns with phrases
  - Find performance on classification tasks, e.g., Syntactic analysis, Semantic analysis, POS tagging, Words sense disambiguation, Name entity recognition, **Topic modeling**, **Intent detection, Hate-speech detection**

- **Interim Results**
  - Innocuous sentences being classified differently
  - How does toxic speech detection performance compare with different types of data?

3. Bias in Models - Findings

- **Implementation and Preliminary Results**

Data Used	F1 score
Unprocessed Data	0.613
Data with recommended disability phrases	0.449
Data with non-recommended disability phrases	0.482
Data that does not mention disability	0.338

Table 1: Comparison of performance of toxic speech detection with three different types of training data

4. What is left? and Plan for Demo

- Bias in **Data**
  - Map of terms of top terms associated with disability data and non-disability data
  - Clustering these terms to categories
- Bias in **Language Representations**
  - System where a user can make a prepared statement based on choices
  - System outputs predicted word and the difference in sentiment score for resultant sentence and a neutral sentence
  - For example, Input: “my deaf latin friend is _____.” Output: <Word>, <Difference>
  - Applying Debiasing
- Bias in **Model**
  - Use different datasets
  - Compare model performance at some or all of the following: Syntactic analysis, Semantic analysis, POS tagging, Words sense disambiguation, **Name entity recognition**, **Topic modeling**, Intent detection, **Hate-speech detection**

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Thank You